

CYBER SECURITY COURSE

Payload Delivery and Reverse Shell Handling

Professional Case Report 14

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Case ID	CSC-2026-14
Discipline	Adversary simulation
Overall risk	Critical
Evidence status	Laboratory methodology documented

Purpose

Understand payload generation, callback configuration, execution visibility and safe session handling.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

A PowerShell-based Meterpreter payload was generated and delivered to a Windows training target, with a Metasploit handler receiving the callback. The lab demonstrated both the effectiveness and visibility of PowerShell payload delivery. Strong script controls and outbound monitoring materially reduce this attack path.

Case Profile

Scenario	A PowerShell-based Meterpreter payload was generated and delivered to a Windows training target, with a Metasploit handler receiving the callback.
Environment	Kali Linux attacker and Windows lab target with direct isolated-network connectivity.
Tools	Veil Evasion, PowerShell, Metasploit, Python HTTP server, Meterpreter

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Configured LHOST and generated a PowerShell reverse-TCP payload.
2. Reviewed the generated batch file and encoded command structure.
3. Started a Metasploit handler using the generated resource file.
4. Hosted the file from the attacker system and executed it on the target.
5. Listed and interacted with the returned session, then removed the payload.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	Encoded PowerShell execution	High	NoProfile, NonInteractive, Hidden and encoded command options reduce user visibility.
F02	Outbound callback	Critical	The target initiates the connection, which can bypass inbound-only controls.
F03	Handler configuration	High	Incorrect LHOST or network segmentation prevents callback and is a common troubleshooting point.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

```
sudo veil
msfconsole -r /var/lib/veil/output/handlers/veil4inpt.rc
python -m http.server
sessions -l
```

```
sessions -i <id>
```

Risk Assessment

Risk	Level	Operational impact
Remote interactive control	Critical	Attacker receives command execution on the endpoint
Detection evasion	High	Encoded and hidden PowerShell may evade basic user awareness

Recommendations

1. Enable PowerShell logging and AMSI-integrated endpoint protection.
2. Block untrusted script execution and unsigned payloads.
3. Inspect unusual outbound callbacks and temporary HTTP servers.
4. Remove generated payloads and terminate handlers after testing.

Knowledge Demonstrated

- Payload generation
- Reverse-shell configuration
- PowerShell analysis
- Session handling

Conclusion

The lab demonstrated both the effectiveness and visibility of PowerShell payload delivery. Strong script controls and outbound monitoring materially reduce this attack path.

Hands On Lab Practical 14

Completed as part of the authorized cyber security course practical work.